

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

II Sem B.Tech. (CBCS) Mid Semester Examinations - I, February 2023

CY1001-1 – ENGINEERING CHEMISTRY

Duration: 1 Hour

Max. Marks: 20

*Note: Answer any **One** full question from **each Unit**.*

	Unit – I	Marks	BT*	CO*	PO*
1. a)	Derive the Nernst equation for the following equilibrium reaction at 298K. $\text{Cu}^{2+} + 2\text{e}^- \rightleftharpoons \text{Cu}$	3	L*2	1	1
b)	Explain the experimental method to determine single electrode potential using calomel electrode.	3	L2	1	1
c)	Describe the construction and working of Lithium-ion battery.	4	L2	1	1
2. a)	What voltage will be generated by a cell that consists of a Fe rod immersed in 0.05 M FeSO ₄ solution and a Pb rod immersed in 0.5 M PbSO ₄ solution at 298 K? Given E° of Fe is -0.44 V and E° of Pb is -0.13 V..	3	L3	1	1
b)	Derive the expression for electrode potential of glass electrode.	4	L2	1	1
c)	Explain the term (i) Current and (ii) Cycle life	3	L2	1	1
	Unit – II				
3. a)	Explain the electrochemical theory of corrosion by taking iron as an example.	5	L2	1	1
b)	What is galvanization? Explain the steps involving in galvanization.	4	L1	1	1
b)	If ship submerged for hundreds of years in sea, it remains almost unaffected. Justify the statement.	1	L3	1	1
4. a)	Explain the corrosion prevention by sacrificial anode and Impressed current method.	5	L2	1	1
b)	Pitting corrosion is a form of accelerated corrosion. Justify the statement.	1	L3	1	1
c)	Describe the construction, working and application of Methanol-Oxygen fuel cell.	4	L2	1	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
